Project Development Phase

Model Performance Test

Date	1 October 2026
Team ID	SWTID-2026-8643
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	10 Marks

Model Performance Testing:

Project team should enter the following information in the model performance testing template.

<table>
<tr><th>S.No.</th><th>Parameter</th><th>Values</th><th>Screenshot</th></tr>
<tr><td>1.</td><td>Metrics</td><td>This project is a role-based access control (ACL) implementation, not a trained ML model. The metrics below are the accessdecision test results for the READ ACL, treating "record visible to a bb1 user" as the positive class (6 records: 2 EEE, 4 ECE/CSE).
Regression Model:
MAE, MSE, RMSE, R2 score – Not applicable (no numeric prediction).
Classification Model:
Confusion Matrix (bb1 user):
<table>
<tr><th></th><th>Pred: Visible</th><th>Pred: Hidden</th></tr>
<tr><td>Actual: EEE</td><td>2 (TP)</td><td>0 (FN)</td></tr>
<tr><td>Actual: ECE/CSE</td><td>0 (FP)</td><td>4 (TN)</td></tr>
</table>
Accuracy Score: 6 / 6 = 100%
Classification Report: Visible class – precision 1.00, recall 1.00, F1 1.00 (support 2). Hidden class – precision 1.00, recall 1.00, F1 1.00 (support 4).
Other users tested: No-role user sees 0 of 6 records (expected 0); admin sees 6 of 6 (expected 6).</td><td>EEE-only list for the bb1 user, empty list for the user without the role, and the admin full list.</td></tr>
<tr><td>2.</td><td>Tune the Model</td><td>Hyperparameter Tuning: Not applicable. The equivalent tuning step was refining the READ ACL script from a basic role check to a layered check: (1) admin gets full access, (2) bb1 gets access limited by the data condition Branch is EEE, (3) everyone else is denied.</td><td>Final ACL script (Advanced Condition) and the delete confirmation screen.</td></tr>
</table>

S.No.	Parameter	Values	Screenshot
		Validation Method: Impersonation testing. Impersonate the EEE user (bb1), a user without the role, and the admin, open u_institution_details.list, and compare the visible records with the expected result. Create, write and delete ACLs were validated the same way with bb1+bb2, bb1+bb2+bb3 and bb1–bb4 roles. **Result after tuning:** Validation accuracy 100%.	